

Arun K Lenin

AI Scientist · LLM Fine-tuning & Agentic Systems · Kaggle Grandmaster

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AI Scientist specializing in LLM fine-tuning, knowledge representation, and agentic AI systems, with production ML infrastructure experience on AWS. Kaggle Grandmaster (Notebooks), ranked #34 globally.

SKILLS

Languages

Python, PySpark, R, SQL, MATLAB, Bash

ML / Deep Learning

LLM fine-tuning, knowledge representation, synthetic data generation, pruning & knowledge distillation, model souping, NER/PII detection, ensemble methods, multimodal VLM fine-tuning, reinforcement learning

NLP & Evaluation

RoBERTa, BERT, RAG, prompt engineering, LLM-as-judge evaluation

Algorithms & Statistics

UMAP, HDBSCAN, Gaussian Processes, submodular optimization, drift detection

Cloud & MLOps

AWS SageMaker, HyperPod, S3, IAM, KMS, Kubernetes, MLflow, distributed training

EDUCATION

B.Tech, Chemical Engineering

Minor: Data Driven Modeling & Systems Engineering
Indian Institute of Technology Madras
2020

RESEARCH

EmbGen: Teaching with Reassembled Corpora

Synthetic data generation algorithm for LLM fine-tuning; captures cross-document concept relationships.
Lead author · arXiv:2605.19394 · under review, EMNLP 2026

COMPETITIVE ML (KAGGLE)

- Grandmaster (Notebooks) - top ~300 globally
- 1st place - Insurance Cross Selling classification
- 2nd place globally - BitGrit NFT Prize Prediction

EXPERIENCE

Commonwealth Bank of Australia, AI Labs

Data Scientist

Jul 2026 - Present

Knowledge representation, LLM fine-tuning, model compression, and agentic AI safety research.

- Customer privacy protection, NER fine-tuning and model compression:** designed a synthetic data generation framework to fine-tune NER models for customer privacy detection; merged fine-tuned model variants using model souping and applied Viterbi decoding with custom loss functions to improve sequence-level accuracy; used structured pruning and knowledge distillation to produce a compact, low-latency model with under 1% performance drop from the teacher model.
- Agentic system user-vulnerability research:** built an end-to-end framework simulating vulnerability scenarios that arise while users interact with agentic systems on a primary task, plus an evaluation pipeline to assess a benchmark dataset of these scenarios; studied frontier LLM behavior under different vulnerable circumstances as part of a responsible AI research effort.

Senior Associate Data Scientist

2024 - Jun 2026

Applied research team: LLM fine-tuning, agentic AI systems, MLOps.

LLM Fine-tuning & Agentic Systems

- Designed EmbGen, a synthetic data generation algorithm capturing cross-document concept relationships; outperformed existing baselines on internal benchmarks. Published as "*EmbGen: Teaching with Reassembled Corpora*" (arXiv:2605.19394), lead author; under review at EMNLP 2026.
- Fine-tuned open-source LLMs for financial-domain tasks, matching or exceeding frontier closed-source model quality on internal evaluations, enabling production deployment of open-source models.
- Designed a semantic drift detection framework for deployed Gen AI agents, combining embedding-based domain representation with LLM-scored retrieval; outperformed standard statistical drift methods.
- Led an AI agent PoC for customer vulnerability identification with an external AI vendor, lifting correct identification from near-zero to 89%.

MLOps & Infrastructure

- Designed a modular OO architecture for LLM fine-tuning, standardized across multiple projects.
- Built an automated end-to-end training pipeline on AWS HyperPod, running pre-training, fine-tuning, inference, and evaluation from a single command with parallel job execution.
- Integrated MLflow for reproducible experiment tracking; managed SageMaker, IAM, KMS, and S3; ran Kubernetes-based distributed training.

Senior Quantitative Model Solutions Specialist

Jul 2022 - Jun 2024

Wells Fargo

- Built a Python tool for automated dashboard and PowerPoint generation of economic forecasts, with a UI for non-technical users - generating hundreds of slides within minutes.
- Supported CCAR and Mid-Year Financial Assessment submissions for the PPNR model across six statistical forecast models; achieved a 90x efficiency gain through automation and integrated PSI monitoring into production controls.

- Four further top-10 finishes
- Notebook held #1 public score on a competition for 2+ years - 19,000+ views, 4,000+ forks

LEADERSHIP

Designed and led a 3-week Gen AI hackathon for 30 graduate hires; mentored junior data scientists; hosted an internal speaker series (Stanford, UNSW).

- Placed 9th globally (out of thousands of entries; 1 of 2 Indian teams in the top 10) in a customer attrition NLP challenge using RoBERTa, BERT, TF-IDF, and ensemble modeling.

Manager 1, Risk Analytics

Oct 2020 - Jun 2022

ICICI Bank

- Developed a master rating scale using K-means clustering for PD bin formation in the retail portfolio.
- Built a statistical profiling model for sales hiring using AHP and Mahalanobis distance on PCA-derived personality traits.
- Automated SME model monitoring (PSI/CSI analysis, WOE recomputation) and validated CECL PD-LGD and ARIMAX models across 21 retail segments.